

Week .3

Course Name : Problem Solving and Testing

Course Code : 10211CS227

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1.Java Date and Time

2.Number of Days Between Two Dates

3.Day of the Year

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1.Java Date and Time

PROGRAM:

```
import java.util.*;

public class Main {

    public static String findDay(int month, int day, int year) {
        Calendar cal = Calendar.getInstance();

        cal.set(year, month - 1, day);

        String[] days = {
            "SUNDAY",
            "MONDAY",
            "TUESDAY",
            "WEDNESDAY",
            "THURSDAY",
            "FRIDAY",
            "SATURDAY"
        };

        return days[cal.get(Calendar.DAY_OF_WEEK) - 1];
    }

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int month = sc.nextInt();
int day = sc.nextInt();
int year = sc.nextInt();

System.out.println(findDay(month, day, year));

sc.close();
}
}
```

OUTPUT:



```
21 05 2006
WEDNESDAY

...Program finished with exit code 0
Press ENTER to exit console.
```

2.Number of Days Between Two Dates

PROGRAM:

```
import java.time.LocalDate;
import java.time.temporal.ChronoUnit;

class Solution {
    public int daysBetweenDates(String date1, String date2) {
        LocalDate d1 = LocalDate.parse(date1);
        LocalDate d2 = LocalDate.parse(date2);

        return (int) Math.abs(ChronoUnit.DAYS.between(d1, d2));
    }
}
```

OUTPUT:

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date1 =
"2019-06-29"

date2 =
"2019-06-30"

Output

1

Expected

1

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date1 =
"2020-01-15"

date2 =
"2019-12-31"

Output

15

Expected

15

3. Day of the Year

PROGRAM:

```
import java.time.LocalDate;

class Solution {
    public int dayOfYear(String date) {
        LocalDate d = LocalDate.parse(date);
        return d.getDayOfYear();
    }
}
```

OUTPUT:

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date =
"2019-01-09"

Output

9

Expected

9

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date =
"2019-02-10"

Output

41

Expected

41

4. Day of the Week

PROGRAM:

```
import java.time.LocalDate;
```

```
class Solution {  
    public String dayOfTheWeek(int day, int month, int year) {  
        LocalDate date = LocalDate.of(year, month, day);  
  
        String dayName = date.getDayOfWeek().toString();  
  
        return dayName.charAt(0) + dayName.substring(1).toLowerCase();  
    }  
}
```

OUTPUT:

Accepted Runtime: 5 ms

✓ Case 1 ✓ Case 2 ✓ Case 3

Input

day =
31

month =
8

year =
2019

Output

"Saturday"

Expected

"Saturday"

Accepted Runtime: 5 ms

✓ Case 1 ✓ Case 2 ✓ Case 3

Input

day =
18

month =
7

year =
1999

Output

"Sunday"

Expected

"Sunday"

Accepted Runtime: 5 ms

✓ Case 1 ✓ Case 2 ✓ Case 3

Input

day =
15

month =
8

year =
1993

Output

"Sunday"

Expected

"Sunday"

5. Java Priority Queue

PROGRAM:

```
import java.util.*;
```

```
class Student {
```

```
    private int id;
```

```
    private String name;
```

```
    private double cgpa;
```

```
    public Student(int id, String name, double cgpa) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        this.cgpa = cgpa;
```

```
    }
```

```
    public int getID() {
```

```
        return id;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
    }
```

```
    public double getCGPA() {
```

```
        return cgpa;
```

```
    }
```

```
}
```

```
class Priorities {
```

```
    public List<Student> getStudents(List<String> events) {
```

```
        PriorityQueue<Student> pq = new PriorityQueue<>(  
            (a, b) -> {
```

```
                if (a.getCGPA() != b.getCGPA()) {
```

```
                    return Double.compare(b.getCGPA(), a.getCGPA());
```

```
                }
```

```
                int nameCompare = a.getName().compareTo(b.getName());
```

```
                if (nameCompare != 0) {
```

```
                    return nameCompare;
```

```
                }
```

```
                return Integer.compare(a.getID(), b.getID());
```

```
            }  
        );
```

```
        for (String event : events) {
```

```
            if (event.startsWith("ENTER")) {
```

```
                String[] parts = event.split(" ");
```

```
                String name = parts[1];
```



```

        double cgpa = Double.parseDouble(parts[2]);
        int id = Integer.parseInt(parts[3]);

        pq.add(new Student(id, name, cgpa));

    } else if (event.equals("SERVED")) {
        if (!pq.isEmpty()) {
            pq.poll();
        }
    }
}

List<Student> result = new ArrayList<>();

while (!pq.isEmpty()) {
    result.add(pq.poll());
}

return result;
}
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

```

```
int n = sc.nextInt();
sc.nextLine();

List<String> events = new ArrayList<>();

for (int i = 0; i < n; i++) {
    events.add(sc.nextLine());
}

Priorities priorities = new Priorities();
List<Student> students = priorities.getStudents(events);

if (students.isEmpty()) {
    System.out.println("EMPTY");
} else {
    for (Student student : students) {
        System.out.println(student.getName());
    }
}

sc.close();
}
```

OUTPUT:

```
5
SURYA 99
VAMSHI 90
SIDDHU 100
HARSHA 91
SIVA 89
EMPTY

...Program finished with exit code 0
Press ENTER to exit console.
```

6.Java Arraylist

PROGRAM:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
ArrayList<ArrayList<Integer>> list = new ArrayList<>();
```

```
// Read the lines
```

```
for (int i = 0; i < n; i++) {
```

```
    int size = sc.nextInt();
```

```
    ArrayList<Integer> row = new ArrayList<>();
```

```
    for (int j = 0; j < size; j++) {
```

```
        row.add(sc.nextInt());
```

```
    }
```

```
    list.add(row);
```

```
}
```

```
// Number of queries
```

```
int q = sc.nextInt();
```

```
for (int i = 0; i < q; i++) {
```

```
    int x = sc.nextInt();
```

```
    int y = sc.nextInt();
```

```
// x and y are 1-based
if (x >= 1 && x <= list.size() &&
    y >= 1 && y <= list.get(x - 1).size()) {

    System.out.println(list.get(x - 1).get(y - 1));

} else {
    System.out.println("ERROR!");
}

}

sc.close();
}
```

OUTPUT:

 Console

 I/O

 AI Agent

5

5 41 77 74 22 44

1 12

4 37 34 36 52

0

3 20 22 33

5

1 3

74

3 4

52

7.Largest Number

PROGRAM :

```
import java.util.*;

class Solution {
    public String largestNumber(int[] nums) {

        String[] arr = new String[nums.length];

        for (int i = 0; i < nums.length; i++) {
            arr[i] = String.valueOf(nums[i]);
        }

        Arrays.sort(arr, (a, b) -> (b + a).compareTo(a + b));

        if (arr[0].equals("0")) {
            return "0";
        }

        StringBuilder result = new StringBuilder();

        for (String s : arr) {
            result.append(s);
        }

        return result.toString();
    }
}
```

```
}  
}
```

OUTPUT :

Accepted Runtime: 1 ms

✓ Case 1 ✓ Case 2

Input

```
nums =  
[10,2]
```

Output

```
"210"
```

Expected

```
"210"
```

Accepted Runtime: 1 ms

✓ Case 1 ✓ Case 2

Input

```
nums =  
[3,30,34,5,9]
```

Output

```
"9534330"
```

Expected

```
"9534330"
```

8. Java Comparator

PROGRAM :


```
import java.util.*;
```

```
class Player {
```

```
    String name;
```

```
    int score;
```

```
    Player(String name, int score) {
```

```
        this.name = name;
```

```
        this.score = score;
```

```
    }
```

```
}
```

```
class Checker implements Comparator<Player> {
```

```
    @Override
```

```
    public int compare(Player a, Player b) {
```

```
        // Sort by decreasing score
```

```
        if (a.score != b.score) {
```

```
            return b.score - a.score;
```

```
        }
```

```
        // If scores are same, sort alphabetically
```

```
        return a.name.compareTo(b.name);
```

```
    }
```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        Player[] players = new Player[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            String name = sc.next();
```

```
            int score = sc.nextInt();
```

```
            players[i] = new Player(name, score);
```

```
        }
```

```
        Arrays.sort(players, new Checker());
```

```
        for (Player p : players) {
```

```
            System.out.println(p.name + " " + p.score);
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

OUTPUT :

```
Console I/O AI Agent
4
surya 500
vamshi 600
siva 700
siddhu 900
siddhu 900
siva 700
vamshi 600
surya 500
```

9.Sort the People

PROGRAM :

```
import java.util.*;
```

```
class Solution {  
    public String[] sortPeople(String[] names, int[] heights) {  
  
        Integer[] index = new Integer[names.length];  
  
        for (int i = 0; i < names.length; i++) {  
            index[i] = i;  
        }  
  
        Arrays.sort(index, (a, b) -> heights[b] - heights[a]);  
  
        String[] result = new String[names.length];  
  
        for (int i = 0; i < names.length; i++) {  
            result[i] = names[index[i]];  
        }  
  
        return result;  
    }  
}
```

OUTPUT :

Accepted Runtime: 1 ms

✓ Case 1

✓ Case 2

Input

names =
["Mary", "John", "Emma"]

heights =
[180, 165, 170]

Output

["Mary", "Emma", "John"]

Expected

["Mary", "Emma", "John"]

Accepted Runtime: 1 ms

✓ Case 1

✓ Case 2

Input

names =
["Alice", "Bob", "Bob"]

heights =
[155, 185, 150]

Output

["Bob", "Alice", "Bob"]

Expected

["Bob", "Alice", "Bob"]